

Maternawell Nigeria Feature Document

Implemented prototype capabilities and current limits | 30 September 2026

Maternawell Nigeria is an offline-capable postnatal depression screening and referral prototype for primary health centres. It supports health workers, facility supervisors, system administrators, and mothers using an anonymous self-check. This document describes the features implemented in the codebase, how the main workflow fits together, and what remains outside the tested prototype. It is intended for the summative demonstration and technical review.

System at a glance

Actor	What the actor can do
Health worker	Sign in, obtain consent, conduct EPDS screenings, review results, confirm urgent safety steps, manage cases and follow-up.
Facility supervisor	Review facility case status, monitor urgent Item 10 flags, acknowledge escalations with notes, and track referral workload.
System administrator	View the facility audit trail and escalation notification queue; no screening write access.
Mother	Use the public anonymous self-check, select a facility, receive guidance and a reference code without a staff account.

Typical case journey

- A health worker signs in and records a mother's informed consent and background details.
- The worker answers ten EPDS items, one question per screen. Draft answers save on the device.
- The app calculates the total and risk tier, then shows a referral plan immediately.
- Any Item 10 score above zero starts a same-day safety and supervisor acknowledgement workflow.
- The case enters the facility list. The worker records contact and referral follow-up.
- When internet access returns, queued changes sync to the server and other authorized facility devices.

Screening and clinical decision support

EPDS questionnaire

The app presents the ten Edinburgh Postnatal Depression Scale items one at a time, with answer choices, progress, and back and next navigation. A screening can be resumed if interrupted. The scoring engine requires an answer from 0 to 3 for every item; the server recalculates the score when a screening is synchronized rather than trusting a client-supplied total.

Total score	Displayed tier	Suggested pathway
0-8	Low	Routine postnatal care, self-care information and community or peer support.
9-12	Moderate	PHC counselling or clinician assessment within one week, plus closer follow-up.
13-30	High	Same-day urgent psychiatric or mental-health specialist referral.
Item 10 > 0	Urgent regardless of total	Same-day safety response and facility supervisor escalation.

Results show the score, tier, suggested actions, and a clear statement that screening is not a diagnosis. The case record retains the plan for later review and PDF export.

Consent and mother information

The staff workflow asks for explicit consent before recording a named screening and captures identifying and postnatal background information. The server rejects screenings without consent. The mother-facing self-check also requires consent, but does not require a name; it can accept optional contact information outside public-demo mode.

Urgent escalation

A nonzero response to Item 10 remains flagged even when the overall EPDS score is low. The health worker sees a required safety-step confirmation and instructions not to leave the mother alone. The facility supervisor sees the unresolved flag, deadline or overdue state, and case context; acknowledgement requires an action note. The flag is not cleared by simply closing the worker view. Urgent cases are queued to the facility dashboard after synchronization. Optional device notifications work while the supervisor dashboard is open; there is no background SMS or push gateway.

Case management and follow-up

The health worker dashboard shows facility screening totals, urgent referrals, moderate-risk cases, open referrals, recent cases, unfinished drafts, and safety confirmations still needed. The case list supports search by name, file number or anonymous reference and filters for risk, follow-up state, source, and sync status.

A case detail page shows the assessment, referral pathway, escalation and sync state, and a follow-up history. Health workers can record pending, contacted, completed, or lost-to-follow-up outcomes with notes. Supervisors can acknowledge urgent flags and can soft-delete a case only with a reason; a pending escalation blocks deletion. Case PDF export is available for authorized use.

Role-specific dashboards and access

Staff sign in with a unique staff ID and password. Online credentials are checked by the Node API, which returns the account's role and facility. The app routes each role to its dashboard and scopes server records to the staff member's facility. A supervisor sees aggregate case status, while item-level answers for ordinary cases are restricted. The administrator has a read-only audit view with action and escalation records.

After a successful online sign-in on a device, the same staff member can unlock their encrypted local data offline with the correct password. The app locks after a reload or ten minutes of inactivity and requires the password again. Demo accounts are provided for the prototype; in ordinary production mode they are not created automatically.

Anonymous mother self-check

The public self-referral flow needs no staff login. It asks the ten EPDS questions, calculates a result, presents stigma-sensitive next steps and nearby facility choices, and issues an anonymous MW reference code. The submission can be queued on the device while offline and later delivered to the selected facility's referral queue. When public-demo mode is enabled, it displays a fictional-data warning and hides the contact-sharing form. This prototype is not monitored for clinical care.

Offline operation and synchronization

- A service worker caches the web app for offline access; the Android package uses the same app through Capacitor.
- Screenings, drafts, audit entries, and an outbox are stored in IndexedDB with encrypted sensitive records.
- The outbox synchronizes on reconnect, at startup, after writes, every 30 seconds while online, and on demand.
- Changes are sent in batches of at most 50, can be gzip-compressed, and are applied idempotently by the server to avoid duplicates.
- Transient failures retry with exponential backoff. Invalid or unauthorized operations are marked failed for inspection. Server changes are pulled by revision cursor, with pagination.
- A sync-status indicator makes pending, successful, and failed synchronization visible to staff.

Security audit and data handling

On-device records use AES-GCM encryption with a key derived from the staff password; the unlocked key lives only in memory. The server stores encrypted SQLite record payloads with AES-256-GCM. Access and refresh credentials are time-limited, and refresh credentials rotate. Protected API endpoints enforce authentication, role, and facility checks. The system records key clinical and administrative actions with actor and time information for the administrator's audit view. Production requires HTTPS termination and a stable server encryption key.

Guidance language and platform

The app includes first-login onboarding, contextual help, a quick-reference guide, a supervisor guide, and troubleshooting content. English is the clinical questionnaire language. Some interface labels have draft Yoruba text, but validated Yoruba EPDS questions are not included. The frontend is a React and Vite progressive web app; a Capacitor Android project targets Android 8.0 and later. The Node API serves the built app and stores synchronized data in SQLite.

Prototype verification and remaining work

As of 30 September 2026, the automated suite passes 18 server tests and 40 app tests. These cover login and roles, scoring boundaries, Item 10 escalation, follow-up, anonymous referral, encryption, offline data and synchronization, and production demo sign-in behind an HTTPS proxy. A production-mode frontend build also succeeds. These checks support the implemented behavior, but do not replace real clinical or field validation.

Area	Current limit
Yoruba EPDS	Validated clinical wording and scoring guidance have not been supplied; the questionnaire remains in English.
Escalation delivery	No background SMS or cross-device push gateway; staff must directly hand off urgent offline cases.
Operations	No in-app account administration or password reset. Public hosting and repository access must be verified separately.
Field readiness	Emergency contact availability, low-end Android use over 2G, and a full 8-12 hour shift have not been validated in the field.

Source and scope

This inventory reflects the current local codebase, the Software Requirements Specification PDF in the repository, the README, and the implementation traceability summary. It describes a functioning prototype and does not certify clinical deployment readiness.